



Figure 5: Edit Toolbar

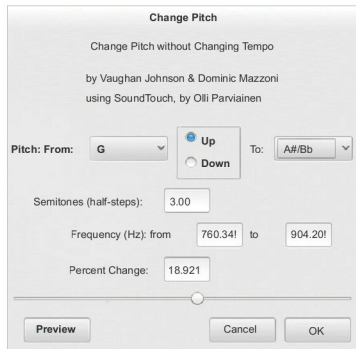


Figure 6: Changing the pitch

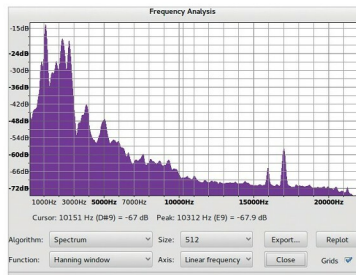


Figure 7: Spectrum of an audio signal

In button. Keep clicking the **Zoom In** button until you see the details.

To delete some part of the audio (like a 15-second clip), follow the steps below:

1. Stop the playback, click near the point where you want the 15-second portion to begin.
2. Zoom in until the **Timeline** shows 15 seconds or more before and after the cursor.
3. Holding down the **Shift** key, click 15 seconds to the right of the cursor.
4. Adjust the start and end of the selection with the mouse.
5. From the **Edit Toolbar**, select the icon 'Trim Audio' or

press the **Delete** key.

6. If you have made a mistake, go to **Edit->Undo**.

Features of Audacity

Audacity has many exciting features. Some of them are discussed here.

Changing the pitch: Select the entire audio (by clicking on the **Track Control** panel). Go to **Effects->Change Pitch**. Now, change the pitch from the given tone to the desired one with the help of the drop-down list. Click **OK**.

Low Pass Filter: This feature of Audacity allows users to keep only those frequencies that are lower than a certain cut-off frequency. It attenuates the frequencies above this threshold. It can be useful to reduce the high-pitched noise. It can also be very helpful for removing vocal parts from the song. Select the track and go to **Effect->Low Pass Filter**.

Alternatively, go to **Effect->LS Filter**. Just change the cut-off frequency and apply.

Equalisation: With the help of this effect, we can manipulate the sounds by frequency. It allows us to increase the volume of certain frequencies and to reduce others. Select the track, go to **Effect->Equalization**.

Let's suppose that in the following example, we want to change the balance of high and low frequencies in the audio, to convert it in the form of a 'Walkie-Talkie'. So from the drop-down menu at the bottom, select the option 'Walkie-Talkie' and click **OK**.

Plot the spectrogram: Select the track. From the drop-down list near the name of the track on the **Track Control** panel, select **Spectrogram**.

Plot the spectrum: Select the track and go to **Analyze->Plot Spectrum**. You can change the window for the plot from the drop-down list.

Audacity is a very useful tool for speech analysis. It is considered the richest tool among all audio analysis software. It is basically good at recording podcasts, different musical tracks, mixing them or separating them, while applying many effects. Moreover, it has the ability to import audio from various sources of recordings, and to export it in digital form. 

References

- [1] <https://github.com/audacity/audacity>
- [2] <http://www.audacityteam.org/download/>
- [3] <https://sourceforge.net/projects/audacity/>
- [4] <http://manual.audacityteam.org/o/index.html>

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